

Distance Z — Testable Primitive (v0.1)

The move this document makes: "distance Z from BEST/SOUND/GOOD" is a symbolic framing until Z is defined as something the math actually produces. This ties Z to a real, already-validated quantity, and states the BSG2 connection as an open hypothesis to check — not a settled equivalence.

Z, defined

$$Z(t) := \alpha(t) = \sup_{\lambda \in \sigma(L(t))} \operatorname{Re}(\lambda)$$

The corrected stability functional from the $V(\psi)$ fix — Pellis's own eq. 69–70, sign-corrected, $\boxed{Z(t) > 0}$ and shrinking toward 0 means the system is losing distance from the collapse threshold. $\boxed{Z(t) = 0}$ is the critical manifold $\boxed{M_c}$ itself. $\boxed{Z(t) \rightarrow 0^-}$ is past it.

This is the same object as $\boxed{\alpha(t)}$ in Prong 1 — no new math, just naming it as the candidate referent for "distance Z," explicitly, so the term stops floating free of the apparatus.

The BSG2 connection — stated as a hypothesis, not asserted

BSG2's BEST/SOUND/GOOD triple-criterion filter is a separate ERES governance construct with its own pass/fail boundary. The claim "distance Z from BEST/SOUND/GOOD" is only checkable once that boundary is expressed in commensurate terms with $\boxed{Z(t)}$. The testable version:

H_Z : BSG2 status transitions toward FAIL as $Z(t) \rightarrow 0$

That's a real, falsifiable correspondence claim — structurally the same kind of bridge hypothesis as $\boxed{H_{\text{bridge}}}$ in Prong 1, between two systems that currently use different vocabularies for what might be the same underlying threshold-distance phenomenon. It has not been checked. It could turn out BSG2's boundary tracks something else entirely (a different axis, not collapse-distance at all), which would be a real and useful finding too.

What this document does not do

It does not validate "Twin-Messiah = 1000-Year SPRT Commitment." That label sits on top of $\boxed{H_Z}$, not inside it — it's the name being given to a hypothesis, not a property the hypothesis has been shown to have. If $\boxed{H_Z}$ is confirmed, the label is still a choice about what to call a confirmed correspondence, not something the confirmation proves.

Status

Open. $\overline{Z(t)}$ is now well-defined. $\overline{H_Z}$ has not been tested against anything — it requires both $\overline{Z(t)}$ (computable in principle from $\overline{\hat{L}(t)}$) and BSG2's actual pass/fail data on the same cases, which doesn't yet exist as a paired dataset.